# The midline of a triangle and of a trapezium

Two theorems with one idea: join the midpoints and you get a segment parallel to the base, of exactly the average length.

LESSON 15 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 10–11 | STAGE 9 · 13.2

LESSON CLOCK

4' | 13' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 4 min |
| Explanation | 13 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- State and prove the midline theorem for a triangle.
- State and use the midline theorem for a trapezium.
- See the triangle case as the trapezium case with $b = 0$.
- Use midlines to solve length problems and to prove other results.

TEXTBOOK

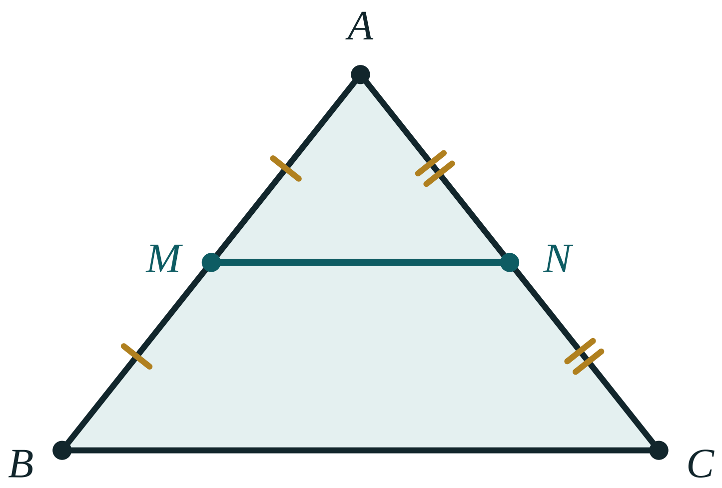
| | |
|---|---|
| National | Темы 10–11, pp. 26–28 |
| Cambridge | Learner's Book pp. 281–284 |
| Workbook | Workbook 13.2 |

## 01 Explanation

### The midline of a triangle

DEFINITION

The **midline** of a triangle is the segment joining the midpoints of two of its sides.




*M and N are the midpoints of AB and AC. The midline MN is parallel to BC and exactly half its length.*

THEOREM

The midline of a triangle is parallel to the third side and equal to half of it:

$$MN \parallel BC \quad \text{and} \quad MN = \frac{1}{2} BC$$

**Proof.** Extend $MN$ beyond $N$ to a point $P$ with $NP = MN$. Then $\triangle ANM \cong \triangle CNP$ by SAS ($AN = NC$, $MN = NP$, vertically opposite angles at $N$). Hence $CP = AM = MB$ and $\angle CPN = \angle AMN$, which are alternate angles — so $CP \parallel MB$. Now $MBCP$ has one pair of sides equal *and* parallel, so by Test 1 it is a parallelogram. Therefore $MP \parallel BC$ and $MP = BC$; and since $MN = \frac{1}{2} MP$, the theorem follows. ∎

CONSEQUENCE

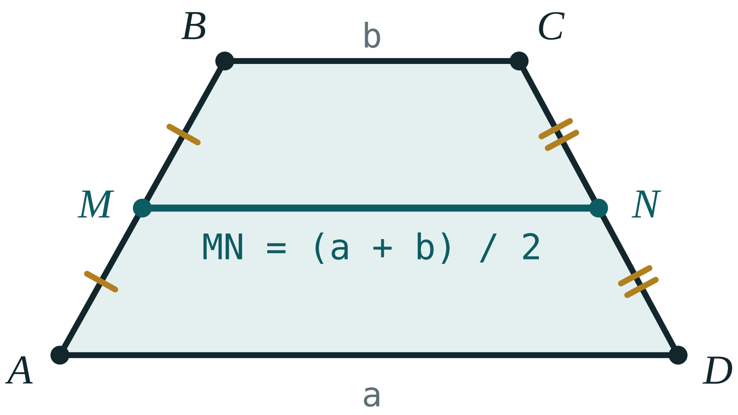
The three midlines of a triangle cut it into four congruent triangles, each similar to the original with half the side lengths and a quarter of the area. The middle triangle has perimeter equal to half the original.

## The midline of a trapezium

THEOREM

The midline of a trapezium — the segment joining the midpoints of the two legs — is parallel to the bases and equal to their half-sum:

$$MN \parallel AD \parallel BC \quad \text{and} \quad MN = \frac{a + b}{2}$$




*The midline of a trapezium is the average of the two bases.*

**Proof.** Draw the diagonal $BD$ and let it meet $MN$ at $K$. In $\triangle ABD$, $MK$ joins the midpoints of $AB$ and $BD$ — a midline, so $MK = \frac{1}{2} AD = \frac{a}{2}$. In $\triangle BCD$, $KN$ is a midline, so $KN = \frac{1}{2} BC = \frac{b}{2}$. Adding gives $MN = \frac{a + b}{2}$. ∎

ONE THEOREM, NOT TWO

Let the shorter base shrink to a point, $b = 0$. The trapezium becomes a triangle and the formula becomes $MN = \frac{a}{2}$ — the triangle midline theorem. The triangle case is the trapezium case with one base of length zero.

## Using midlines

Two habits that solve most midline problems:

- When you see **two midpoints**, join them — you have created a midline.
- When you see **one midpoint** and want a second, draw a parallel to a side; the midpoint corollary from the last lesson supplies it.

Area is also useful: the area of a trapezium is $S = MN \cdot h$ — the midline multiplied by the height.

## 02 Worked examples

**EXAMPLE 1** ***In $\triangle ABC$, $BC = 18$ cm. $M$ and $N$ are the midpoints of $AB$ and $AC$. Find $MN$.***

1. $MN$ joins two midpoints — it is a midline.

2. $MN = \frac{1}{2} BC$

   Midline theorem.

3. $MN = 9\ cm$

**ANSWER**

$MN = 9\ cm$

**EXAMPLE 2** **A trapezium has bases 8 cm and 14 cm and height 5 cm. Find its midline and area.**

1. $MN = \frac{8 + 14}{2} = 11\ cm$

   Half-sum of the bases.

2. $S = MN \cdot h = 11 \cdot 5$

   Area of a trapezium.

3. $S = 55\ cm^2$

**ANSWER**

$MN = 11\ cm, S = 55\ cm^2$

EXAMPLE 3

***The midline of a trapezium is 12 cm and one base is 3 cm shorter than the other. Find both bases.***

1. $a + b = 2 \cdot 12 = 24$

   From the midline formula.

2. $a = b + 3$

   The given relation.

3. $b + 3 + b = 24 \implies 2b = 21$

4. $b = 10.5,\ a = 13.5$

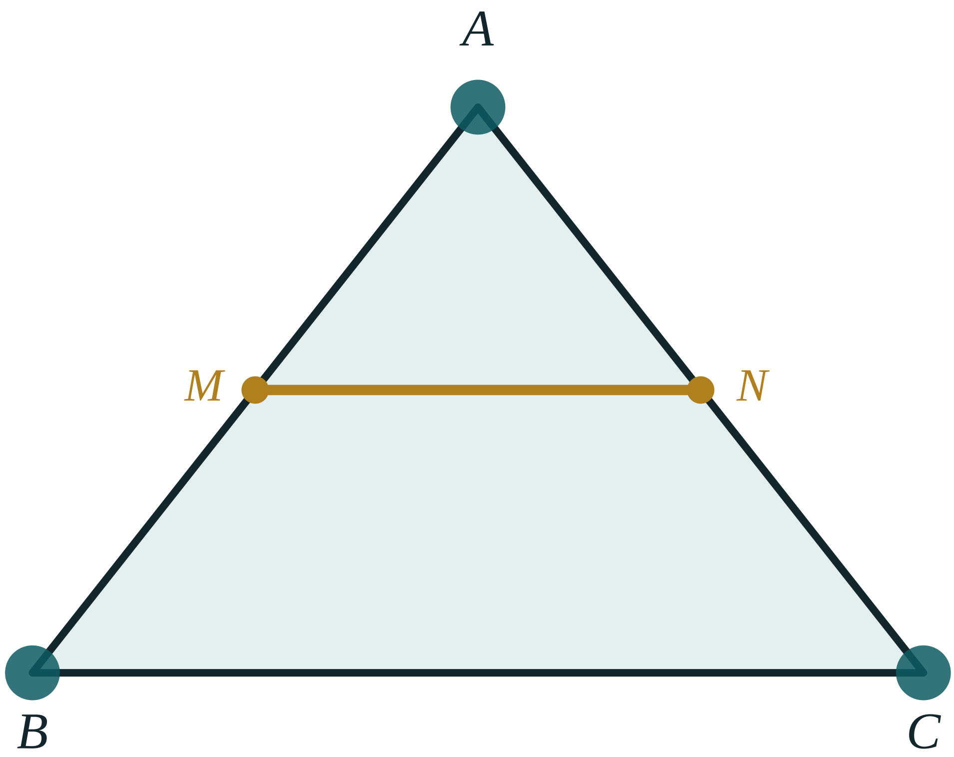
ANSWER

$10.5\ cm$ and $13.5\ cm$

## 03 Interactive model

Drag a vertex to a wild position — the ratio BC : MN stays at 2.00 throughout.

**The midline of a triangle**




BC **260** | MN **130** | BC / MN **2**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Midline of a triangle | Uchburchakning o‘rta chizig‘i | Средняя линия треугольника |
| Midline of a trapezium | Trapetsiyaning o‘rta chizig‘i | Средняя линия трапеции |
| Midpoint | O‘rta nuqta | Середина |
| Half-sum | Yarim yig‘indi | Полусумма |
| Corollary | Natija | Следствие |
| Congruent triangles | Teng uchburchaklar | Равные треугольники |
| Parallel | Parallel | Параллельный |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

## 05 Quick check

### Check yourself

The midline of a triangle with base 20 cm is:

- *40 cm*
- *10 cm*
- *20 cm*
- *5 cm*

The midline of a trapezium with bases 7 and 13 is:

- *20*
- *10*
- *6*
- *91*

The three midlines of a triangle cut it into:

- 2 triangles
- 3 triangles
- 4 congruent triangles
- 6 triangles

A trapezium has midline 9 and height 6. Its area is:

- *54*
- *27*
- *15*
- *108*

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up

7 PROBLEMS

1. *In $\triangle ABC$, $BC = 18$ cm. Find the midline $MN$ parallel to $BC$.*

   ANSWER *9 cm*

2. *A midline of a triangle is 7 cm. Find the side it is parallel to.*

   ANSWER *14 cm*

3. *A trapezium has bases 6 cm and 10 cm. Find the midline.*

   ANSWER *8 cm*

4. *A trapezium has bases 5 cm and 15 cm. Find the midline.*

   ANSWER *10 cm*

5. *A trapezium has midline 9 cm and height 6 cm. Find the area.*

   ANSWER *54 cm²*

6. *A triangle has sides 10, 12, 14. Find the perimeter of the triangle formed by its midlines.*

   ANSWER *18*

7. *The midline of a trapezium is 11 cm and one base is 8 cm. Find the other.*

   ANSWER *14 cm*

## Medium · the standard question

7 PROBLEMS

1. *A trapezium has bases 8 cm and 14 cm and height 5 cm. Find the midline and area.*

   ANSWER *11 cm, 55 cm²*

2. *The midline of a trapezium is 12 cm and one base is 3 cm shorter than the other. Find both.*

   ANSWER *10.5 cm* and *13.5 cm*

3. *In $\triangle ABC$ the midlines have lengths 4, 5, 6. Find the sides of $\triangle ABC$.*

   ANSWER *8, 10, 12*

4. *A trapezium has area 60 $cm^2$ and height 5 cm. Find its midline.*

   ANSWER *12 cm*

5. *The midline of a trapezium is 10 cm and the bases are in the ratio 2 : 3. Find them.*

   ANSWER *8 cm* and *12 cm*

6. *In $\triangle ABC$, $M$ and $N$ are the midpoints of $AB$ and $AC$, and $MN = 6.5$ cm. Find $BC$.*

   ANSWER *13 cm*

7. *A trapezium has bases 9 and 21 and legs 10 and 10. Find the midline and the height.*

   ANSWER Midline *15*; $\frac{21-9}{2} = 6$, so $h = 8$

## Hard · stretch and reason

7 PROBLEMS

1. *Prove the midline theorem for a triangle.*

   ANSWER Extend $MN$ to $P$ with $NP = MN$; $\triangle ANM \cong \triangle CNP$ by SAS makes $MBCP$ a parallelogram, so $MP = BC$ and $MN$ is half of it.

2. *Prove that the midline of a trapezium equals the half-sum of the bases.*

   ANSWER The diagonal $BD$ splits $MN$ into two triangle midlines of lengths $\frac{a}{2}$ and $\frac{b}{2}$.

3. *The midlines of a triangle cut it into four triangles. Compare their areas with the original.*

   ANSWER Each is $\frac{1}{4}$ of the original area, and all four are congruent.

4. *The midline of a trapezium is 14 cm and the bases are in the ratio 3 : 4. Find both bases.*

   ANSWER $a + b = 28$ in the ratio $3 : 4$: $12$ cm and $16$ cm.

5. *P, Q, R, S are the midpoints of the sides of any quadrilateral. Prove PQRS is a parallelogram.*

   ANSWER $PQ$ and $SR$ are both midlines parallel to the diagonal $AC$ and equal to half of it.

6. *The midline of a triangle divides it into a triangle and a trapezium. Compare their areas.*

   ANSWER The small triangle is $\frac{1}{4}$ of the whole, so the trapezium is $\frac{3}{4}$ — a ratio of $1 : 3$.

7. *A trapezium has midline 8 and the two bases differ by 4. Find the area if the height is 7.*

   ANSWER Bases $6$ and $10$; $S = 8 \cdot 7 = 56$.

## 07 Homework

### Homework — 5 problems

Geometry 8, Темы 10–11, pp. 26–28.

1. *In $\triangle ABC$, $BC = 22$ cm. Find the midline parallel to $BC$.*
2. *A trapezium has bases 11 cm and 17 cm. Find the midline, then the area if the height is 6 cm.*

3. *The midline of a trapezium is 15 cm and one base is 4 cm longer than the other. Find both.*
4. *The midlines of a triangle are 6, 7 and 8. Find the perimeter of the triangle.*
5. *Prove that the midline of a triangle is parallel to the third side.*